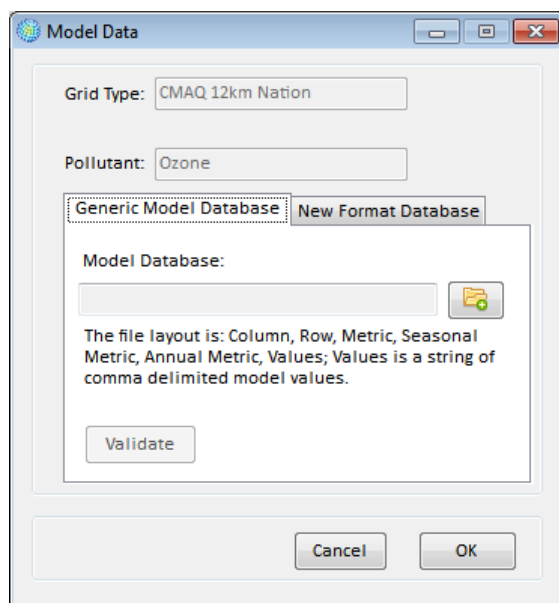


5.1 Model Data

After choosing the *Model Data* option, use the **Generic Model Database** tab to load grid-definition-based modeling data (e.g., CMAQ or CAMx).¹⁹ For model files being imported from an Excel workbook with multiple tabs, BenMAP-CE will prompt you to select the tab containing the data for import.



The **Model Database** specifies the location of the air quality model results that you want to import. Table 5-1 presents the structure that these files must have, and Table 5-2 presents a sample data file with a variety of metrics. (For more information on air quality models, the EPA website has detailed descriptions of a variety of models at <https://www.epa.gov/scram>.)

Table 5-1. Air Modeling Data File Variables

Variable	Type	Required	Notes
Column	Integer	Yes	The column and the row uniquely identify each set of modeling values and link the modeling data with cells in a Grid Definition.
Row	Integer	Yes	
Metric	Text	No	This variable is either blank (signifying that the Values are Observations, rather than Metric values), or must reference an already defined Metric (e.g., 1-hour daily maximum) for the appropriate Pollutant.

¹⁹ Community Multi-scale Air Quality (CMAQ) Model is available at: <http://www.epa.gov/amad/Research/RIA/cmaq.html> or <https://www.cmascenter.org/cmaq/>. Comprehensive Air Quality Model with Extensions (CAMx) is available at: <http://www.camx.com/>.

Seasonal Metric	Text	No	This variable is either blank (signifying that the Values are not Seasonal Metric values) or must reference an already defined Seasonal Metric for the Metric (e.g., mean of the 1-hour maximum values for the months of June through August).
Annual Metric	Text	No	This variable is either blank (signifying that the values are not an annual metric) or must be one of: <i>None</i> , <i>Mean</i> , <i>Median</i> , <i>Max</i> , <i>Min</i> , <i>Sum</i> (e.g., mean of the 1-hour maximum values for the year)
Values	Comma Separated Values (Text) ²⁰	Yes	If Metric is blank, values are supplied as a comma-delimited string of values for the year [e.g., 365 or 366 (leap year) values for daily data, 8760 or 8784 (leap year) values for hourly data]. If Metric is defined, but Seasonal Metric and Annual Metric are blank, 365 or 366 metric values. If Seasonal Metric is defined, but Statistic is blank, <i>n</i> seasonal metric values. If Annual Metric is defined, one annual statistic value for either the Metric (if Seasonal Metric is blank) or the Seasonal Metric. Missing values are signified with a period ('.').

Table 5-2. Sample Air Modeling Data File

Column	Row	Metric	Seasonal Metric	Annual Metric	Values
24	101				,5.90,,,11,,,
24	102				19.80,,,3.5,,,14.70,,,,,4.90
24	103				11.5,,,1.5,,,10.20,,,2.5,,,2.6
24	104				12.60,,,4.60,,,7.30,,,,,9.20
24	105	DailyAverage			,5.90,,,11,,,
24	106	DailyAverage			19.80,,,3.5,,,14.70,,,,,4.90
24	107	DailyAverage			11.5,,,1.5,,,10.20,,,2.5,,,2.6
24	108	DailyAverage			12.60,,,4.60,,,7.30,,,,,9.20
24	109	DailyAverage	QuarterlyAverage		11.82,13.24,18.79,14.25
24	110	DailyAverage	QuarterlyAverage		12.31,13.89,18.27,21.19
25	101	DailyAverage	QuarterlyAverage		12.33,15.68,18.49,15.96
25	102	DailyAverage	QuarterlyAverage		13.52,13.69,19.04,22.43
25	103	DailyAverage		Mean	14.52
25	104	DailyAverage		Mean	16.41
25	105	DailyAverage		Mean	15.62
25	106	DailyAverage		Mean	17.17
25	107	DailyAverage	QuarterlyAverage	Mean	14.29
25	108	DailyAverage	QuarterlyAverage	Mean	18.97
25	109	DailyAverage	QuarterlyAverage	Mean	20.14
25	110	DailyAverage	QuarterlyAverage	Mean	16.46

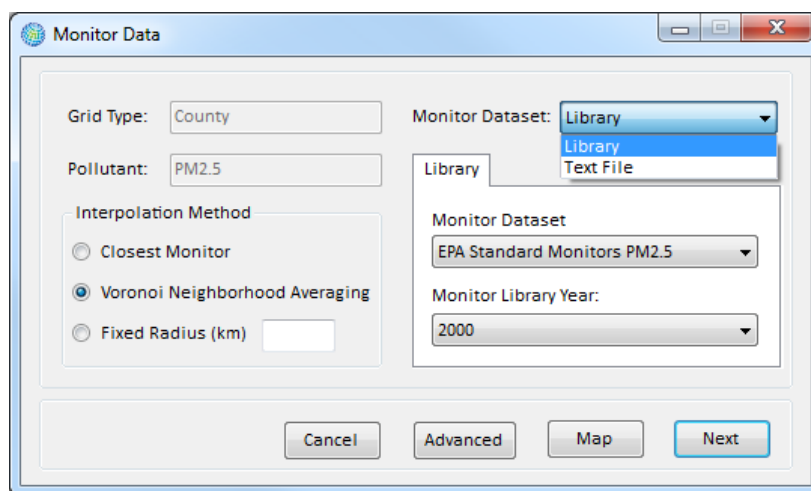
²⁰ The list of comma-separated values must be surrounded by quotation marks, otherwise only the first value in the list will be used.

5.2 Monitor Data

Using the *Monitor Data* grid creation option, you create an air quality grid directly from air pollution monitoring data. At the top left of the **Monitor Data** window, you will see the previously selected grid definition in the **Grid Type** field, and the previously selected pollutant in the **Pollutant** field. Below the **Pollutant** field of the **Monitor Data** window, you are asked to select an **Interpolation Method**. The interpolation method is used to move from point-based monitor data to grid cell-based air quality data. That is, some grid cells will have many monitors in them, some will have just one, and some will have none. BenMAP-CE uses the interpolation methods to generate representative air quality metric values for each grid cell from monitor data for all of these cases.

Fundamental Concept - Interpolation

Interpolation is the process of estimating the air quality level in an unmonitored area using data from one or more nearby air quality monitors. BenMAP-CE includes three types of interpolation options: one assigns the value from the closest monitor; a second assigns the weighted average of all monitors within a user-specified fixed radius, and the third applies the Voronoi Neighbor Averaging method. These interpolation methods are discussed in more detail below and in Appendix B.



BenMAP-CE includes three **Interpolation Methods**. The *Closest Monitor* method simply uses the monitor closest to a grid cell's center as its representative value. The *Voronoi Neighborhood Averaging* method takes an inverse-distance weighted average of a set of the monitors surrounding a grid cell's center as its representative value. The *Fixed Radius* method averages all of the monitors within a fixed (user-specified) radius measured from the center of the grid cell. Each method is described below. For more detail, also see Appendix B on Air Pollution Exposure Estimation Algorithms.

On the right side of the **Monitor Data** window, you can specify a source for your monitor data. Using the **Library** tab, you may select from data that you have already loaded into BenMAP-CE. Choose the **Monitor Dataset** and **Monitor Library Year** from the drop-down lists.

If you want to load your own monitor data, choose *Text File* from the **Monitor Dataset** drop-down list. You can then browse to locate the data file you want to load. See Chapter 4 for more information about formatting monitor datasets.

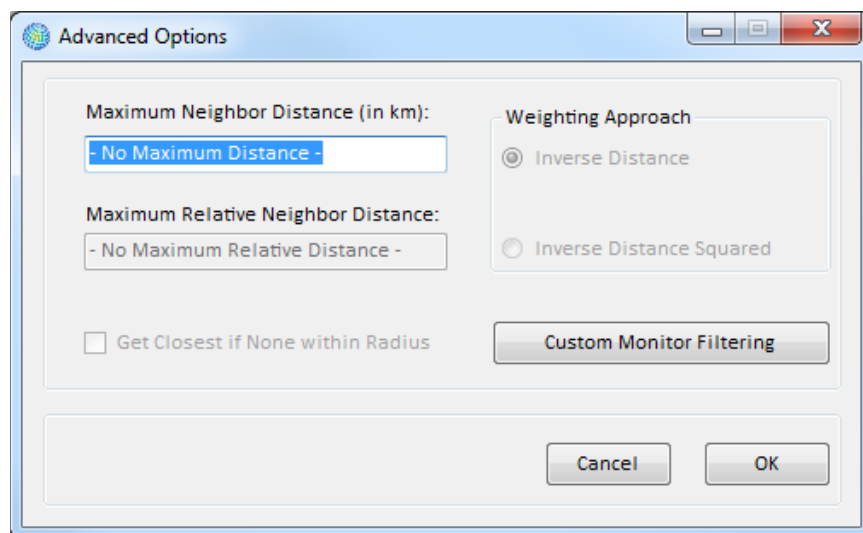
At the bottom of the **Monitor Data** window, there is a **Map** button. The **Map** button opens the **Monitor Map** window, allowing you to preview the map that you are about to load.

Decision Point – Monitor Data

Monitor data may be stored as part of a BenMAP-CE setup *or* loaded as a text file for a particular analysis. If you have already loaded your monitor data into a setup, choose *Library* from the **Monitor Dataset** drop-down list and select the appropriate monitor data in the **Library** tab. If you want to load your own monitor data, choose *Text File* from the **Monitor Dataset** drop-down list and browse to locate the data file you want to load.

5.2.1 Closest Monitor for Monitor Data

If you choose the *Closest Monitor* option, BenMAP-CE identifies the monitor closest to each grid cell's center, and then assigns that monitor's data to the grid cell. *Closest Monitor* interpolation can be modified by clicking on the **Advanced** button at the bottom of the window and typing in a **Maximum Neighbor Distance (in km)**.



The **Maximum Neighbor Distance** specifies the maximum distance (measured in kilometers) that a monitor may be from the center of a grid cell. Cells without any monitors within this distance will not be included in the resultant air quality grid. The default setting is infinite (i.e., no limit to the monitor's distance from the center of the grid cell).

Note: The **Maximum Relative Neighbor Distance** and the **Weighting Approach** options are irrelevant (and are therefore disabled) when using the *Closest Monitor* method, since BenMAP-CE is only choosing a single monitor to assign to any given grid cell.